When Does Task-Conditioned Cognitive Configuration Search Generalize? A Controlled Audit of LLM Agent Scaffolds with Fair Baselines, Token Costs, and a Cross-Benchmark Test

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Abstract

LLM agents are usually deployed with a fixed, hand-designed cognitive configuration: a reasoning strategy, an episodic memory/exemplar policy, and a context budget chosen once and reused across tasks. We ask a narrower and more auditable question than "can we evolve better agents": do task-conditioned choices of these configuration components materially change accuracy and inference cost above a frozen backbone, and does automatic search over them find configurations whose advantage survives fair baselines and unseen data? We study this with a three-tier audit of QA-style single-agent scaffolds over frozen Qwen2.5 backbones, keeping every tier's evidence status explicit. **Tier 1** (previously locked, descriptive): on an untouched GSM8K holdout, a search-selected chain-of-thought (CoT) configuration outperformed four fixed baselines that all used direct answering, by +29.0 percentage points at 1.5B and +16.7 points at 7B—an architecture-matters result that confounds search with the CoT-vs-direct gap and cannot be attributed to search alone. **Tier 2** (fair-baseline dev audit): against CoT-matched presets on the reused GSM8K dev slice, the search-selected configuration shows no statistically detectable accuracy advantage over the strongest preset (60.0% vs 59.0%, $+1.0$pp, $p{=}1.0$); its +10-point edge over a no-memory CoT control is exploratory ($p{=}0.087$). **Tier 3** (pre-locked cross-benchmark confirmation): on 300 locked SVAMP test items never used for search or tuning, the search-selected configuration scores 70.33%, with no statistically detectable accuracy difference from either the strong CoT preset (71.67%; primary contrast -1.33pp, 95% CI $[-6.00, +3.33]$, exact McNemar $p{=}0.67$) and the no-memory control (70.33%; 0.00pp, $p{=}1.0$)—and the dev-side memory benefit does not replicate. Cost analysis is equally blunt: the no-memory control matches the searched configuration's observed accuracy with $1.75\times$ fewer total tokens. Equal-budget evolutionary search also does not beat random search in our compact audited space, and a correctly implemented trainable Mamba-2 substrate is not selected as beneficial. Our contribution is not a new optimizer but a falsifiable audit protocol—fair CoT-matched baselines, item-level paired statistics, total-token cost accounting, and a pre-locked cross-benchmark confirmation—together with its two-sided findings.

1 Introduction

Assembling an LLM agent involves choices that are usually made by hand and then frozen: whether to reason directly or step by step, whether to carry an episodic memory of exemplars and in what form, and how much of the context budget to spend on them. A growing line of work automates these choices—searching agent programs, modular designs, or multi-agent topologies—and typically reports that the searched artifact beats the manual one on the benchmark used for searching. What is audited far less often is whether such advantages survive two elementary controls: (i) baselines matched on the *components* the search actually varies, and (ii) evaluation on data the search never touched.

This paper reports such an audit for task-conditioned *cognitive configuration search*: per-task selection of a reasoning strategy, an episodic memory/exemplar policy, and a context allocation policy above a frozen backbone. Our setting is deliberately modest—QA-style single-agent scaffolds, not interactive tool-using or multi-agent systems, and not neural architecture search over backbone weights—because the audit questions are already sharp there, and because leakage-safe paired evaluation is feasible at that scale.

We organize the evidence into three tiers with explicitly different epistemic status (Figure 1). **Tier 1** is a previously locked holdout study (frozen

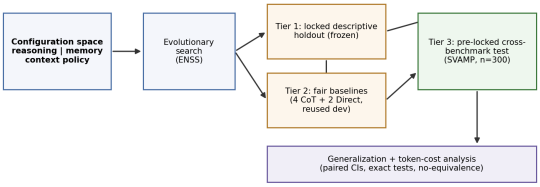


Figure 1: Audit overview. A structured cognitive configuration space (reasoning, memory, context) is searched evolutionarily; selected configurations are then audited in three tiers of increasing evidentiary strength—a frozen descriptive holdout (Tier 1), fair component-matched baselines on reused dev (Tier 2), and a pre-locked cross-benchmark single-shot test (Tier 3)—followed by joint generalization and token-cost analysis.

before this paper): its headline—a searched configuration beating the strongest fixed baseline by +29.0 percentage points at 1.5B and +16.7 at 7B on GSM8K holdout—is real but descriptive, because all four fixed baselines used direct answering while the searched configuration used CoT. **Tier 2** repairs baseline fairness: six predefined baselines (four CoT-matched and two Direct controls) compared against the search-selected configuration on the same dev slices the search used (exploratory by construction). **Tier 3** is a pre-locked confirmation: three configurations frozen before any inference, evaluated once on the 300-item original SVAMP test split, a public benchmark never used for search, calibration, or tuning.

Contributions. We report three scientific findings and one methodological contribution. (1) *Apparent search gains can be largely explained by reasoning-scaffold differences.* The locked +29.0/+16.7pp holdout gap appears only against direct-answering baselines; once baselines are CoT-matched, the search-selected configuration shows no statistically detectable accuracy advantage (60.0% vs 59.0% on reused dev; 70.33% vs 71.67% on the locked SVAMP test). (2) *Task-conditioned configurations show limited cross-benchmark transfer.* The dev-side memory benefit of the GSM8K-selected configuration (+10.0pp, exploratory) does not replicate on the pre-locked SVAMP test (0.00pp): configuration optimization is task-dependent, and dev-side selection signals are not self-certifying. (3) *Accuracy improvements must be analyzed jointly with inference cost.* On the locked SVAMP test, the no-memory control matches the searched configuration's observed accuracy (70.33%) with 1.75× fewer total tokens; accuracy-only reporting would miss this cost structure entirely. (4) *An auditable evaluation protocol.* CoT-matched baselines, item-level paired statistics with pre-specified contrasts, byte-hashed pre-locked cross-benchmark evaluation, and full prompt+completion token accounting—a checklist that makes configuration-search claims falsifiable. Throughout, we keep the negative results on the record: equal-budget evolutionary search does not outperform random search on the enumerated landscape, and a trainable Mamba-2 memory substrate is not selected as beneficial.

2 Related Work

Automated agent design. ADAS (Hu et al., 2024) searches over agent programs expressed in code; AgentSquare (Shang et al., 2025) searches a modular design space of planning, reasoning, tool-use, and memory modules; MaAS (Zhang et al., 2025) performs query-dependent multi-agent architecture search through an agentic supernet and explicitly prices inference resources. These systems optimize performance (MaAS also cost) on the search distribution. We differ in target and emphasis: our search target is the internal cognitive configuration of a single frozen-backbone agent, and our main object of study is the gap between search-side selection and held-out, fair-baseline, cross-benchmark evidence—including where the search procedure itself fails to beat random search. Evo-Memory (Wei et al., 2025) and related work study continually evolving agent memory during deployment; here memory is not evolved online—the memory policy is one component of a task-conditioned configuration searched against a fixed, calibration-only experience bank. DSPy (Khattab et al., 2023) compiles declarative LLM programs into optimizable pipelines; EvoPrompt (Guo et al., 2023) evolves prompt text. We treat reasoning strategy as one gene jointly audited with memory and context policy.

Scaffolds and prompting. ReAct (Yao et al., 2023), Reflexion (Shinn et al., 2023), and chain-of-thought prompting (Wei et al., 2022) are hand-designed cognitive configurations in our terms; verification and planning variants appear as reasoning genes. Retrieval-augmented generation (Lewis et al., 2020) grounds LLMs in external evidence; our memory policies manage a leakage-safe, calibration-only exemplar bank, and

we quantify when such memory helps, hurts, or is neutral.

Search methodology. Our evolutionary loop follows regularized evolution (Real et al., 2019) and NSGA-II-style non-dominated sorting with crowding distance (Deb et al., 2002); Bayesian optimization is the standard sample-efficiency reference (Snoek et al., 2012); AlphaEvolve (Google DeepMind, 2025) shares the evolutionary loop but evolves programs. Unlike NAS over backbone weights, our search operates above a frozen LLM, and we enumerate the compact space exhaustively so evolutionary search can be audited against equal-budget random search on a known landscape. Mamba (Gu and Dao, 2023) and Mamba-2 (Dao and Gu, 2024) provide the selective state-space models behind our trainable memory substrate, which is implemented and searchable but—as we report—not selected as beneficial.

Evaluation. We use GSM8K (Cobbe et al., 2021), SVAMP (Patel et al., 2021), PubMedQA (Jin et al., 2019), and QASPER (Dasigi et al., 2021) in its LongBench form (Bai et al., 2023) with frozen Qwen2.5-Instruct backbones (Yang et al., 2024). Across these lines of work, evaluation typically centers on the search distribution itself; whether discovered configurations yield transferable cognitive improvements or benchmark-specific specialization—and at what inference cost—remains insufficiently studied. Our audit targets exactly this gap. We claim no priority over any of the above; the contribution is the audit protocol and its two-sided findings.

3 Setting and Audit Design

Cognitive configuration space. A configuration is a structured genome over three gene families: (i) *memory policy*—none / recency / retrieval (term-frequency cosine or hashed bag-of-words cosine over a calibration-only exemplar bank) / a trainable Mamba-2 substrate / hybrid, with recall width and exemplar-count sub-genes; (ii) *reasoning strategy*—direct, chain-of-thought (with depth), verify, planner; (iii) *context policy*—full, truncated, or answer-only exemplar verbosity with word budgets. The trainable Mamba-2 substrate is included as one candidate configuration family among several, allowing us to test whether architectural memory mechanisms provide measurable benefits; it is not itself a contribution of this paper. Normalization removes semantically inactive sub-genes before hashing so duplicate phenotypes are never double-counted. A compact audit layer (memory $\times$ reasoning $\times$ context $= 48$ configurations) is fully enumerated per benchmark as a ground-truth landscape.

Three evidence tiers. **Tier 1—locked descriptive holdout (frozen).** Pre-specified, commit-locked holdout evaluation of task-selected configurations vs four fixed baselines (all direct-answering) on GSM8K/PubMedQA/QASPER, plus the exhaustive-landscape search-efficiency audit; locked before this paper and reported with confounds made explicit. **Tier 2—fair-baseline dev audit.** Six predefined baselines (four CoT-matched and two Direct controls) matched to the search-selected configuration's reasoning strategy, plus the search-selected REF itself, run on the same dev slices the search used (GSM8K test[0:100], QASPER items[0:50]) and a pre-chosen 7B shortlist; exploratory, with provenance gaps (§5). **Tier 3—pre-locked cross-benchmark confirmation.** Three configurations (search-selected REF, strongest CoT preset B2, no-memory CoT control B1) frozen with all 300 SVAMP test IDs, file hashes, model revision, and analysis plan *before any inference*; evaluated exactly once; primary and secondary contrasts fixed in advance. Independent of the project's selection history, but a public benchmark (pretraining contamination cannot be ruled out) and not same-distribution with GSM8K.

Statistics. All key comparisons are paired at the item level with paired bootstrap 95% CIs (10,000 resamples, fixed seed). For binary outcomes (GSM8K, SVAMP, PubMedQA), significance uses two-sided exact McNemar tests; for continuous QASPER item-level F1, a two-sided exact paired sign test over non-tied differences. Absence of significance is not treated as equivalence.

Scope. These are QA-style single-agent cognitive scaffolds above a frozen LLM. We do not evaluate interactive tool-using, multi-agent, embodied, or long-horizon environment agents, and our conclusions do not directly extend to them.

4 Tier 1 — Locked Descriptive Holdout: Architecture Matters, Search Credit Unclear

The frozen pre-study (pre-specified, commit-locked before holdout inference) task-searched configurations on dev slices and evaluated once on untouched holdouts (GSM8K test[100:1319]; PubMedQA 400 items; QASPER 100 items; Qwen2.5-1.5B/7B).

Headline, with its confound. On GSM8K holdout the search-selected configuration (recency memory, CoT depth 3, full context) reaches 0.518 at 1.5B and 0.859 at 7B, beating the strongest fixed baseline by +29.0 and +16.7 percentage points (McNemar $p<0.001$). However, all four fixed baselines answered *directly*; the searched configuration used CoT. The gap is therefore best read as "cognitive architecture choices matter materially," not as isolated evidence for search or for memory: Tier 2 shows CoT-matched presets close most of it.

Search-side selection. Task-conditioned preferences were stable and interpretable: GSM8K consistently favored CoT-style reasoning across seeds (3/3, depth 2–3) with the lightweight memory choice varying among none/recency/retrieval; PubMedQA converged bit-identically to no-memory direct answering; QASPER shifted the optimum toward retrieval memory with many exemplars—a pre-specified prediction that was confirmed. But dev-selected task-specific configurations did not universally generalize: on PubMedQA and QASPER holdout they were beaten by the GSM8K-selected configuration.

Search-efficiency audit (negative). On the fully enumerated compact landscape (48 configurations), online evolutionary search does not beat equal-budget uniform random search at any budget $\{12, 24, 36, 48\}$ on any benchmark (20 seeds; differences below 0.01 in AUC and hypervolume). We make no optimizer-superiority claim.

Substrate audit (negative). A real, trainable Mamba-2 memory substrate participates in the search but is not selected as beneficial. Controlled same-genome ablations found memory effects to be task- and scale-dependent (neutral at 1.5B on GSM8K, +12.6pp at 7B; +18.0pp at 1.5B on PubMedQA; +1.5pp on QASPER); memory causality is claimed only where the comparison is controlled.

Config	Memory / reasoning	Acc (%)	Prompt tok
B0	none / direct	11.0	6,791
B5	recency×1 / direct	13.0	23,091
B1	none / CoT d3	50.0	8,091
B4	recency×1, 128w / CoT d3	49.0	14,491
B3	retrieval tf-cos / CoT d3	57.0	27,683
B2	recency×3 / CoT d3	59.0	67,391
REF	recency×1 / CoT d3	**60.0**	24,391

Table 1: GSM8K dev (n=100, 1.5B), accuracy and prompt-token cost for the six predefined baselines and the search-selected REF. REF shows no statistically detectable advantage over the strongest CoT-matched baseline B2 (+1.0pp, exact McNemar p=1.0); REF−B1 = +10.0pp (95% CI $[0.0, 20.0]$, p=0.087, exploratory). Direct baselines (B0/B5) collapse: the Tier-1 headline confounds reasoning strategy with search.

Extraction caveat. A QASPER configuration that works at 1.5B shows near-zero extracted-answer F1 at 7B; a frozen diagnostic reveals perfect marker compliance with the answer placed before the marker—a backbone–prompt–extraction interaction with a substantial extraction component. It is not evidence of a pure capability collapse, and the diagnostic does not establish that underlying answer quality is unaffected.

5 Tier 2 — Fair CoT-Matched Baselines on Reused Dev (Exploratory)

Tier 1's fairness gap motivated baselines matched on the searched configuration's actual components. We evaluated six predefined baselines (four CoT-matched: B1–B4; two Direct controls: B0/B5) plus the search-selected REF on GSM8K dev (n=100) and QASPER dev (n=50) at 1.5B, with a pre-chosen 7B shortlist (0.16 GPU-hours total; Table 1).

GSM8K dev. Among CoT configurations REF shows no statistically detectable accuracy difference from B2 or B3 (REF−B3 +3.0pp, p=0.63). The reduced-exemplar-verbosity baseline B4 was lower than REF on this exploratory dev slice (+11.0pp, unadjusted p=0.043, secondary); a single contrast on reused dev cannot rank factor importance, so causal prioritization of context budget over preset choice requires further controls. The REF−B1 gap reflects the memory+exemplar component under CoT, not search;

at 7B (dev, secondary, unadjusted) it is +10.0pp (p=0.041), directionally consistent with Tier-1's scale-dependent memory ablation but exploratory here.

QASPER dev (exploratory, extraction-sensitive). REF (0.167) is slightly *below* B1/B3/B2 (0.184–0.192); paired sign tests are non-significant. Raw model outputs for these cells were not preserved by the legacy evaluator—an evidence limitation we carry.

Provenance limitations. The F1 run manifest covers only the last four cells (overwrite-on-invocation defect) and records the runner git SHA as "unknown" (VM checkout lacked `.git`); per-item prediction files for all 15 cells are committed and were independently re-scored, but exact runner-source provenance is partially undocumented. F1 numbers are accepted-with-limitations, not paper-grade confirmation.

6 Tier 3 — Pre-Locked SVAMP Confirmation (Single Shot)

Protocol. Before any inference we committed: all 300 original SVAMP test IDs in order with file SHA-256s (dataset revision pinned); the three configurations (REF, B2, B1); the backbone (Qwen2.5-1.5B-Instruct, exact revision); the Body+Question adapter and numeric scorer (versioned, unit-tested on synthetic fixtures; Equation/Answer never enter prompts); the primary contrast (REF vs B2) and secondary (REF vs B1); and the analysis (paired bootstrap CI + exact McNemar). Memory exemplars came from the GSM8K *train* calibration bank only; SVAMP train was never read. Zero normalized exact-match overlap between the 300 locked items and the full GSM8K corpus was verified pre-inference. A post-hoc read-only forensic audit re-hashed every committed artifact byte-for-byte (dataset, lock, 900 predictions, paired statistics) and passed, with two honestly annotated gaps: the run-time loader had checked ordered IDs and count but not full file bytes (closed post-hoc by the audit), and the on-VM model-revision observation is not itself a committed artifact.

Results. One shot, 0 errors, 0.06 GPU-hours (Table 2, Figure 2). The search-selected REF shows **no detectable advantage** over the strong preset B2; B2 is nominally +1.33pp with 2.04× REF's total tokens. B1 matches REF's observed accuracy with 1.75× fewer total tokens (3.56× fewer prompt tokens): in the observed accuracy–token plane REF is dominated by B1.

Config	Acc (%)	Prompt	Compl.	Total
REF	70.33	68,017	50,049	118,066
B2	**71.67**	197,017	44,137	241,154
B1	70.33	**19,117**	48,264	**67,381**

Table 2: Locked SVAMP test (n=300, 1.5B), single shot. Accuracy in percent (REF 211/300, B2 215/300, B1 211/300) with prompt, completion, and total tokens. Primary: REF−B2 = −1.33pp, 95% CI $[-6.00, +3.33]$, exact McNemar p=0.67. Secondary: REF−B1 = 0.00pp, CI $[-5.33, +5.33]$, p=1.0. B1 matches REF's observed accuracy with 1.75× fewer total tokens—an observation, not statistically demonstrated dominance.

Non-transfer of the dev-side memory benefit. Tier-2's REF−B1 point estimate (+10.0pp on GSM8K dev) was *not replicated* on SVAMP (0.00pp; Figure 3). We state this precisely: the dev estimate did not replicate cross-benchmark. This is not a formal demonstration of a significant task×memory interaction (contrasting p-values is not an interaction test), not evidence of equivalence, and not proof that memory never transfers—it is one paired, pre-locked non-replication at n=300 resolution (±∼5pp).

7 What the Evidence Supports

Supported. (1) *Cognitive configuration choices materially affect accuracy and cost*: direct vs CoT reasoning swings GSM8K dev accuracy 11–13% → 49–60%; the locked holdout gap of the searched CoT configuration over direct fixed baselines is +29.0/+16.7pp; CoT variants span ∼8.3× in prompt tokens with observed accuracies of 49–60%, so the cost–accuracy tradeoff is configuration-dependent. (2) *Task-conditioned selection is stable and interpretable at the family level*: CoT-dominant configurations win on GSM8K across seeds; PubMedQA converges to no-memory direct; QASPER shifts toward retrieval memory (pre-specified, confirmed). (3) *Search-side preferences can fail to generalize*: dev-selected task-specific configurations lost to the GSM8K configuration on two holdouts (Tier 1); the dev-side memory benefit did not replicate on SVAMP (Tier 3). Held-out, fair-baseline, cross-benchmark auditing is what separates specialization from generalizable advantage.

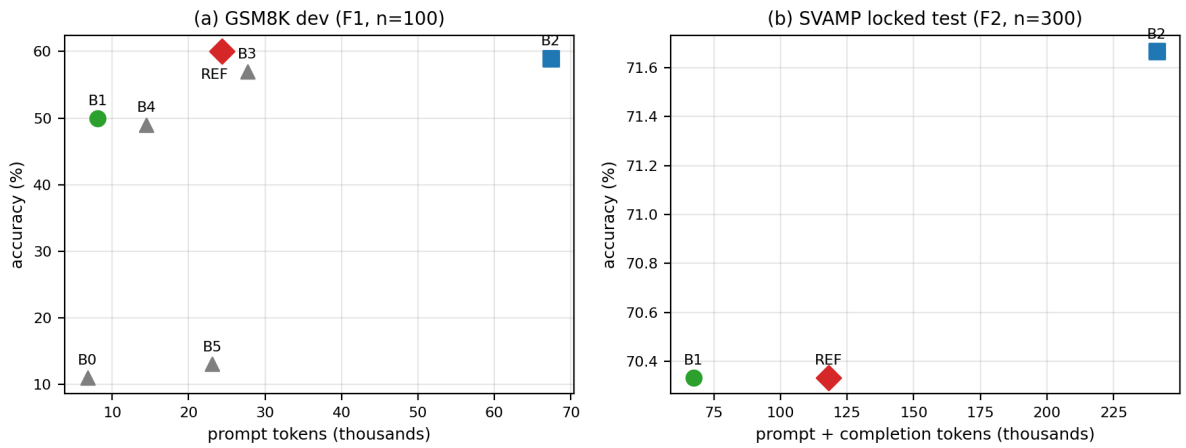


Figure 2: Accuracy vs. inference-token cost, separate panels per dataset. (a) GSM8K dev (F1, n=100, prompt tokens): direct baselines collapse; CoT configurations cluster at 49–60% with $\sim$8.3$\times$ prompt-token spread; REF shows no detectable advantage over B2 at mid-range cost. (b) Locked SVAMP test (F2, n=300, prompt+completion tokens): B1 attains REF's observed accuracy at the lowest cost; B2 is nominally highest in both.

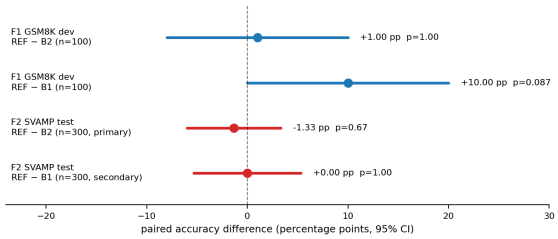


Figure 3: Paired accuracy differences with 95% bootstrap CIs (blue: F1 GSM8K dev; red: F2 locked SVAMP test). CIs are sampling uncertainty, not equivalence bounds: absence of a detectable difference does not establish that the compared configurations are equal.

Unsupported (claimed nowhere). (1) *Search superiority*: no detectable REF advantage over strong presets in Tiers 2–3; evolutionary search $\approx$ equal-budget random search on the enumerated landscape (Tier 1). (2) *Universal or scale-growing memory benefit*: effects are task- and scale-dependent; the one pre-locked cross-benchmark test found none. (3) *Neural substrate benefit*: the trainable Mamba-2 substrate is implemented and searchable but not selected as beneficial.

8 Reproducibility and Ethics

All evaluation code, locked protocols, configuration files, per-item predictions (900/900 with full raw completions for Tier 3), statistics scripts, the read-only forensic verifier, and figure scripts are included in the anonymous supplementary. Tier-1 artifacts are frozen at their locked commits; Tier 3 was pre-locked before inference and is auditable byte-for-byte. Tiers 2–3 together used $<$ 0.25 GPU-hours. The work uses public benchmarks and public models, involves no human subjects, and reports negative results and failure modes explicitly. We release no new model weights. The public models and benchmarks we use carry known limitations—benchmark contamination, annotation noise, and the general misuse potential of LLMs—and our released configurations and predictions inherit them; we do not claim our audit protocol eliminates these risks, and downstream users should scrutinize reused artifacts accordingly.

9 Limitations

Single model family (Qwen2.5, 1.5B/7B). SVAMP is a public benchmark—pretraining contamination cannot be ruled out, and it is not same-distribution with GSM8K, so Tier 3 is cross-benchmark rather than i.i.d. confirmation. Dev slices are small (n=100/50) and were exposed to selection (Tier 2 is exploratory; 7B and B4 contrasts are secondary and unadjusted for multiplicity). n=300 gives $\pm\sim$5pp CI resolution; absence of detectable differences is not equivalence. Tier-1 fixed baselines were direct-only, so its headline cannot be attributed to search alone. F1 provenance is partially undocumented (manifest coverage, runner git SHA) and QASPER dev raw outputs were not preserved. QASPER scoring

is extraction-sensitive (marker placement), which we diagnose but do not claim to fix. Our scaffolds are QA-style single-agent configurations; nothing here directly addresses interactive tool-use, multi-agent, embodied, or long-horizon agents, or backbones outside one family.

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